NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	09/01/2026	Issued For Review	MHP	AS	MR		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

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Rev.	Tanggal	Deskripsi

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skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR FLAME DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 5 dari 21

Tag Number	: See List	Location	: See List
P&ID No.	: -	Equip./Line No.	: -

GENERAL	1	Tag No.	See List
	2	Service	See List
	3	Sensor Type	UV/IR type Detector
	4	Sensor Casing	Stainless Steel
	5	Area Classification	Zone 2
	6	Enclosure Certification	IP65 & Explosion Proof
	7	Mounting	Bolted swivel mounting
	8	Power Supply	24 VDC
	9	Enclosure Material	Stainless Steel / Copper Free Aluminium
	10		
DETECTOR	11	Transmitter output	4-20 mA
	12	Signal For;	Signal For; 0 mA DC Fault
	13		2 mA DC Sensor Fault
	14		4 mA DC Normal Condition
	15		8 mA DC IR activated
	16		12 mA DC UV activated
	17		16 mA DC Warning
	18		20 mA DC Alarm Signal
	19	Accuracy	± 0.3 mA
	20	Monitoring System	Microprocessor system
	21	Response time	< 3 seconds @ 50 ft
	22	Sensitivity	Max 80 ft distance for 1 sq.ft (0.092 m2) Gasoline Fire
	23	Cone Vision	120 degree
	24	System check	Continuous Optical Path Monitoring
	25	Status indication	Two LED-RED (Alarm), GREEN (Normal Oper.),
	26		GREEN-Blinking (Fault) (2)
	27	Fault diagnosis	Required
	28	Range	UV pass band : 185 to 260 nm; IR center wavelength: 4.35 Microns
	29	Electrical connection	½" NPTF
	30	Power supply unit	Required, 24 VDC © 2Amp
	31	Wiring termination	3 wires (1 Triad)
ENVIROMENT SPECIFICATION	32	Ambient Temperature	27.2 °C (min. 25.5 °C-max. 29 °C)
	33	Relative Humidity	82.1 % (min. 74 %- max. 90.3 %)
	34	Barometer	1 atm
	35		
OPTIONS	36	Special tools	Test lamp/Test Kit (4)
	37		
SELECTION	38	Manufacture	As per AML / PML
	39	Model No	By VENDOR (2)

Notes:

1. Vendor to provide 316 Stainless Steel name plate / tag number permanently attached.
2. VTA = Vendor to advise & complete.
3. All necessary calibration equipment to be supplied with detectors covered in this datasheet.
4. Vendor to supply special tools for calibration.
5. Detectors shall be provided with suitable test connections for calibration.
6. Vendor to supply 316SS mounting bracket & accessories.

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DATA SHEET FOR FLAME DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 6 dari 21	

FLAME DETECTOR LIST

No.	Tag Number	Location	Service	P&ID No.	REMARK
1	FD-001	FIELD AREA	T-003	TMBP-SP001-IN-LAY-D108	OUTDOOR
2	FD-002	FIELD AREA	T-003	TMBP-SP001-IN-LAY-D108	OUTDOOR
3	FD-003	FIELD AREA	P-002 A/B	TMBP-SP001-IN-LAY-D108	OUTDOOR
4	FD-004	FIELD AREA	DEG-001/2	TMBP-SP001-IN-LAY-D108	OUTDOOR
5	FD-005	FIELD AREA	T-006A	TMBP-SP001-IN-LAY-D108	OUTDOOR
6	FD-006	FIELD AREA	T-0O5A	TMBP-SP001-IN-LAY-D108	OUTDOOR

skkmigas PERTAMINA EP	**PERTAMINA EP REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR GAS DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 7 dari 21

Tag Number	: See List	Location	: See List
P&ID No.	: -	Equip./Line No.	: -

GENERAL	1	Tag No,	See List	
	2	Application	Detection of hydrocarbon gas	
	3	Sensor Type	Infra Red Point Type Detector	
	4	Enclosure Class	Explosion Proof, Eex"d", IP 65	
	5	Mounting	2" Pipe Mounting	
	6	Enclosure Material	Coper-free Aluminium / Stainless Steel	
	7	Area Classification	Zone 2, Group HA, T3	
	8			
DETECTOR	9	Transmitter Output	4 - 20 mA (LOOP HART ■ SMART)	
	10	Detection Range	0-100% LEL	
	11	Accuracy	3% of full scale range 0-50% LEL, 5% of full scale for 50-100% LEL	
	12	Response Time	50% < 2 Secs & 90% < 4 Secs (VTA)	
	13	System Checks	Self Diagnostic Test	
	14	Alarm	Set Point: 20% LEL (High), 60% LEL(High-High)	
	15		Fault: 0 mA	
	16		Cal & Zero: 1.5 mA	
	17		Zero Reading: 4 mA ± 0.2 mA	
	18		0 to 100 % LEL: 4-20 mA	
	19	Fault Diagnostic	Required	
	20	Calibration	Note	(3)
	21	Power Supply	24 VDC @0.4Amp (MFTR STD)	
	22	Electrical Connections	½" NPTF	
	23	Wiring/Termination	3 wires (1 Triad)	
	24			
OPTIONS	25	Calibration Equipment	Note	(4) & (5)
	26	Calibration Test Conn.	Note	(6)
	27	Gas Sampling	Required	(2)
	28	Local Display	YES	
	29	Mounting Bracket	Required	(7)
	30			
ENVIRONMENT SPECIFICATION	31	Ambient Temperature	27.2 °C (min. 25.5 °C-max. 29 °C)	
	32	Relative Humidity	82.1 % (min. 74 %- max. 90.3 %)	
	33	Barometer	1 atm	
	34			
SELECTION	35	Manufacture	As per AML / PML	(8)
	36	Model No	By VENDOR	(2)

Notes:

1. Vendor to provide 316 SS nameplate or tag number permanently attached to instrument
2. VTA: Vendor to advise & complete.
3. Detector shall be calibrated for hydrocarbon gas and shall be able to eliminated any interference from rain, fog, mist or undirect sunlight & lightning flashes.
4. All necessary calibration equipment to be supplied with detectors and shall include a portable gas containing 50% LEL mixture of Methane (CH4)
5. Vendor to supply special tools for calibration.
6. Vebdor shall advice and provide Detectors with suitable test connections for any required calibrator.
7. Vendor to supply 316SS mounting bracket & accessories for each tag.
8. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)

skkmigas PERTAMINA EP	**PERTAMINA EP REGIONAL 1 ZONA 4**	PGG
DATA SHEET FOR GAS DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 8 dari 21	

GAS DETECTOR LIST-SP1

No.	Tag Number	Location	Service	P&ID No.	REMARK
1	GD-001	FIELD AREA	PRODUCTION SEPARATOR(V-001 & V-002)	TMBP-SP001-IN-LAY-D108	OUTDOOR
2	GD-002	FIELD AREA	TEST SEPARATOR (V-003 & V-004)	TMBP-SP001-IN-LAY-D108	OUTDOOR
3	GD-003	FIELD AREA	PIG RECEIVER	TMBP-SP001-IN-LAY-D108	OUTDOOR
4	GD-004	FIELD AREA	TEST TANK TMB(T-003)	TMBP-SP001-IN-LAY-D108	OUTDOOR
5	GD-005	FIELD AREA	TEST TANK KRG (T-004)	TMBP-SP001-IN-LAY-D108	OUTDOOR
6	GD-006	FIELD AREA	TEST PUMP KRG (P-003 A/B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
7	GD-007	FIELD AREA	TEST PUMPTMB (P-002 A/B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
8	GD-008	FIELD AREA	WASH TANKTMB (T-001)	TMBP-SP001-IN-LAY-D108	OUTDOOR
9	GD-009	FIELD AREA	WASH TANK KRG (T-002)	TMBP-SP001-IN-LAY-D108	OUTDOOR
10	GD-010	FIELD AREA	GAS ENGINE GENERATOR (GEG-001 & GEG-002)	TMBP-SP001-IN-LAY-D108	OUTDOOR
11	GD-011	FIELD AREA	KOD PUMP (P-006 A/B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
12	GD-012	FIELD AREA	CLOSED DRAIN PUMP (P-007 A/B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
13	GD-013	FIELD AREA	STORAGE TANK TMB(T-005B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
14	GD-014	FIELD AREA	GROSS TRANSFER PUMP (P-001 A/B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
15	GD-015	FIELD AREA	STORAGE TANK TMB (T-005A)	TMBP-SP001-IN-LAY-D108	OUTDOOR
16	GD-016	FIELD AREA	STORAGE TANK KRG (T-006B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
17	GD-017	FIELD AREA	STORAGE TANK KRG (T-006A)	TMBP-SP001-IN-LAY-D108	OUTDOOR
18	GD-018	FIELD AREA	NCLOSURE GAS ENGINE GENERATOR(GEG-001 & GEG-002	TMBP-SP001-IN-LAY-D108	OUTDOOR

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR SMOKE DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 9 dari 21

Tag Number	: See List	Location	: See List
Drawing No.	: -	Equip./Line No.	: -

GENERAL	1	Quantity	See List	
	2	Detector Type	Addressable Ionization Smoke Detector	
	3	Hazardous Area Classification	Non Hazardius Area / Safe Area	
	4	Enclosure Protection Type	IP 42	
	5	Required Certification	UL, FM	
	6	Application	Designed to sense both visible and invisible products of combustion	
	7	Installation	Indoor	
	8	Construction	High Impact Fire Retardant Plastic	
	9	Mounting	Yes, Direct Mounting to Ceiling	
	10			
SENSOR SPECIFICATION	11	Sensor type	Ionization Type	
	12	Coverage Single Detector	9,1 mtr	
	13	Response Time	2.5% / ft	
	14	Range Min / Max	32° F to 120°F	
	15			
ELECTRIC SPECIFICATION	16	Power Supply	24 VDC (by Fire Alarm Control Panel)	
	17	Power Consumption	Mfr. Std.	(4)
	18	Output Signal	Digital	
	19	Contact Rating	1A @ 24 VDC	
	20	Cable Type Required	14 to 22 AWG	
	21	Electrical Connection	½" NPT	
	22	No. of terminal	4 (Four)	
	23	LED	Yes	
	24			
ENVIRONMENT SPECIFICATION	25	Ambient Temperature	67.64°F to 98.96°F	
	26	Relative Humidity	35% to 100% RH	
	27	Barometer	14.63 to 14.66 Psi	
	28			
OTHER	29	Sensitivity Setting	Adjustable at site	
	30	Detector Fault and Health Indication	Yes	
	31	Factory & Test Calibration	Yes	
	32	Test Tools	Note	(6)
	33			
REMARKS	34	Name Plate (Tagging Number)	YES, 316SS	(1)
	35	Manufacture	As per AML / PML	(2)
	36	Model No.	By VENDOR	(3)

Notes:

1. All smoke detector shall be tagged with 316 SS nameplate showing tag no.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. VENDOR shall submit proper model number on bid proposal.
4. As per manufacture standard
5. If malfunction is detected, the output current is rised to fault level and fault relay is energized.
6. Vendor to supply special tools for calibration.

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DATA SHEET FOR SMOKE DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 10 dari 21	

SMOKE DETECTOR LIST

No.	Tag Number	Location	Service	P&ID No.	REMARK
1	SD-001	FIELD AREA	ELECTRICAL PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
2	SD-002	FIELD AREA	ELECTRICAL PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
3	SD-003	FIELD AREA	BATTERY ROM	TMBP-SP001-IN-LAY-D109	INDOOR
4	SD-004	FIELD AREA	INSTRUMENT PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
5	SD-005	FIELD AREA	INSTRUMENT PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
6	SD-006	FIELD AREA	OPERATOR ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
7	SD-007	FIELD AREA	DOCUMENT ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
8	SD-008	FIELD AREA	MANAGER ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
9	SD-009	FIELD AREA	LOBBY AREA	TMBP-SP001-IN-LAY-D109	INDOOR
10	SD-010	FIELD AREA	FRONT OFFICE & LOBBY AREA	TMBP-SP001-IN-LAY-D109	INDOOR
11	SD-011	FIELD AREA	FRONT OPERATOR ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
12	SD-012	FIELD AREA	MEETING ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
13	SD-013	FIELD AREA	MEETING ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
14	SD-014	FIELD AREA	LABORATORY	TMBP-SP001-IN-LAY-D109	INDOOR
15	SD-015	FIELD AREA	PAINTRY	TMBP-SP001-IN-LAY-D109	INDOOR
16	SD-016	FIELD AREA	WAREHOUSE	TMBP-SP001-IN-LAY-D109	INDOOR
17	SD-017	FIELD AREA	WAREHOUSE	TMBP-SP001-IN-LAY-D109	INDOOR
18	SD-018	FIELD AREA	GEG-01	TMBP-SP001-IN-LAY-D109	INDOOR
19	SD-019	FIELD AREA	GEG-02	TMBP-SP001-IN-LAY-D109	INDOOR
20	SD-020	FIELD AREA	DEG-01	TMBP-SP001-IN-LAY-D109	INDOOR

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR HEAT DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 11 dari 21

Tag Number	: See List	Service	: See List
P&ID No.	: -	Equip./Line No.	: -

GENERAL	1	Detector	Fixed Temperature Detector	
	2	Quantity	See List	
	3	Area Classification	Non Hazardous Area / Safe Area	
	4	Ingress Protection	Min IP42	
	5	Required Certification	UL, CSFM, FFM	
	6	Installation	Indoor	
	7	Enclosure Material	Fiber Glass / Plastic Enclosure	
	8			
HEAT DETECTOR	9	Item to be Detected	Heat	
	10	Type	Addressable	
	11	Fix Temperature	58 °C	(4)
	12	Good Maintainability	Yes	
	13	Sensitivity	Note	(4)
	14	Output Signal	Digital	
	15	Power Supply	24 VDC, Loop Powered	
	16	Accuracy	Note	(4)
	17	Indicator	Yes, Red	
	18	Electrical Connection	½" NPT	
	19			
ENVIRONMENT SPECIFICATION	20	Ambient Temperature	27.2 °C (min. 25.5 °C-max. 29 °C)	
	21	Relative Humidity	82.1 % (min. 74% - max. 90.3%)	
	22	Rainfall	14.9 mm (min. 0.5 mm - max. 64.8 mm)	
	23	Velocity	2.11 m/s (min. 1.5 m/s - max. 3.2 m/s)	
	24	Barometer	1 atm	
	25			
OTHER	26	Mounting	Required, As per Manufacturer's Standard	
	27	Factory & Test Calibration	Yes, Required	(5)
	28	Accessories	End Of Line (EOL) Resistor	
	29			
REMARKS	30	Name Plate (Tagging Number)	YES, 316SS	(1)
	31	Manufacture	As per AML / PML	(2)
	32	Model No.	By VENDOR	(3)
	33			
	34			
	35			
	36			
	37			
	38			
	39			
	40			

Notes:

1. All detector shall be tagged with 316SS nameplate showing tag no.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. VENDOR shall submit proper model number on bid proposal
4. VTA = Vendor to Advise
5. Vendor to supply special tools for calibration.

HEAT DETECTOR LIST

No.	Tag Number	Location	Service	P&ID No.	REMARK
1	HD-001	FIELD AREA	BATTERY ROM	TMBP-SP001-IN-LAY-D109	INDOOR
2	HD-002	FIELD AREA	ELECTRICAL PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
3	HD-003	FIELD AREA	ELECTRICAL PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
4	HD-004	FIELD AREA	INSTRUMENT PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
5	HD-005	FIELD AREA	INSTRUMENT PANEL ROOM	TMBP-SP001-IN-LAY-D109	INDOOR
6	HD-006	FIELD AREA	OPERATOR ROOM	TMBP-SP001-IN-LAY-D109	INDOOR

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	POG

DATA SHEET FOR LINE HEAT DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 13 dari 21

Tag Number	: See List	Location	: See List
Drawing No.	: -	Equip./Line No.	: -

GENERAL	1	Detector Type	Line Heat Detector	
	2	Quantity	See List	
	3	Area Classification	Non Hazardous Area / Safe Area	
	4	Required Certification	FM, UL	
	5	Installation	Indoor & Outdoor	
	6			
SENSOR SPECIFICATION	7	SensorType	Sensor Cable	
	8	Sensor Material	Two Steel Conductor	
	9	Function	Switch, Alarm	
	10	Setting	Adjustable	
	11	Measuring Range	32 - 302 °F	
	12	Wire Configuration	2 Wire	
	13	Measuring Junction	Yes	
	14	Seath Diameter	4 mm or Mfr. Std	
	15	Element Length	30 ft / 9 m	(4)
	16	Insulation Material	Heat Sensitive Polymer	
	17	Protective Tape	Mfr. Std.	
	18	Outer Covering	Polymer Jacket, 356 °F	
	19	Resistance Value	0 - 100 kΩ (tolerance 1%)	
	20	Accuracy	1%	
	21	Clips, Drive Rings, Beam clamps,	Required	
	22			
SWITCH	23	Switch Type	Snap, Dry Contact, Hermetically Sealed	
	24	Switch Form	SPDT	
	25	Enclosure	IP 5	
	26	Power Supply	24 VDC	
	27	Accuracy	1% FS	
	28	Adjustable Range	32 - 302 °F	
	29	Set Point	248 °F (adjustable)	
	30	Electrical Connection	½" NPTF	
	31	Color	Red	
	32	Body Material	Aluminium Finished Std.	
	33	Mounting	yes c/w mounting bracket	
	34			
ENVIRONMENT SPECIFICATION	35	Ambient Temperature	67.64 °F to 98.96 °F	
	36	Relative Humidity	35% to 100% RH	
	37	Barometer	14.63 to 14.66 Psi	
	38			
REMARKS	39	Name Plate (Tagging Number)	YES, 316SS	(1)
	40	Manufacture	As per AML / PML	(2)
	41	Model No.	By VENDOR	(3)

Notes:

1. All Heat Detector shall be tagged with 316 SS nameplate showing tag no.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. VENDOR shall submit proper model number on bid proposal.
4. VENDOR shall be advised sensor element length requirement.
5. Wiring Box and accessories shall be supplied by VENDOR
6. Vendor to supply special tools for calibration.

skkmigas PERTAMINA EP	**PERTAMINA EP REGIONAL 1 ZONA 4**	PGG
DATA SHEET FOR LINE HEAT DETECTOR	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 14 dari 21	

LINE HEAT DETECTOR LIST

No.	Tag Number	Location	Service	P&ID No.	REMARK
1	LHD-002	FIELD AREA	TEST TANK TMB (T-003)	TMBP-SP001-IN-LAY-D108	OUTDOOR
2	LHD-003	FIELD AREA	TEST TANK KRG(T-004)	TMBP-SP001-IN-LAY-D108	OUTDOOR
3	LHD-004	FIELD AREA	WASH TANK TMB (T-001)	TMBP-SP001-IN-LAY-D108	OUTDOOR
4	LHD-005	FIELD AREA	WASH TANK KRG (T-002)	TMBP-SP001-IN-LAY-D108	OUTDOOR
5	LHD-006	FIELD AREA	INSTRUMENT PANEL ROOM	TMBP-SP001-IN-LAY-D108	OUTDOOR
6	LHD-009	FIELD AREA	STORAGE TANK TMB (T-005B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
7	LHD-011	FIELD AREA	STORAGE TANK TMB (T-005A)-EXISTING	TMBP-SP001-IN-LAY-D108	OUTDOOR
8	LHD-012	FIELD AREA	STORAGE TANK KRG (T-006B)	TMBP-SP001-IN-LAY-D108	OUTDOOR
9	LHD-013	FIELD AREA	STORAGE TANK KRG (T-006A)	TMBP-SP001-IN-LAY-D109	OUTDOOR

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	POG

DATA SHEET FOR MANUAL CALL POINT (INDOOR)	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 15 dari 21

Tag Number	: See List	Location	: See List
Drawing No.	: -	Equip./Line No.	: -

GENERAL	1	Tag No.	See List	
	2	Application	Manual Initiation of Fire Alarm	
	3	Area Classification	Safe Area	
	4	Enclosure Class	IP42	
	5	Mounting	Surface	
	6	Enclosure Material	Glass reinforced polyester / GRP	(4)
	7	Quantity	See List	
	8			
SPECIFICATION	9	Type	Break glass	
	10	Enclosure Colour	Red	
	11	Push Button Colour	Red or Manufacture Standard	
	12	Protective Cover	Required	
	13	Contact Form	2 (two) SPDT (VTA)	
	14	Power Supply	24 VDC @ 1 Amp	
	15	Cable Entry	½" NPTF	
	16	Connection	Screwed	
	17	Wiring / Termination	2 wires	
	18			
ENVIRONMENT SPECIFICATION	19	Ambient Temperature	27.2 °C (min. 25.5 °C - max. 29 °C)	
	20	Relative Humidity	82.1 % (min. 74% - max. 90.3%)	
	21	Rainfall	14.9 mm (min. 0.5 mm - max. 64.8 mm)	
	22	Velocity	2.11 m/s (min. 1.5 m/s - max. 3.2 m/s)	
	23	Barometer	1 atm	
	24			
OPTIONS	25	Mounting Bracket	YES	(4)
	26	Cover	Cover shall be provided to prevent mis-operation	
	27	LED Indicator	N/A	
	28	Resistor	YES	(5)
	29	Reset Key	YES	
	30			
SELECTION	31	Name Plate (Tagging Number)	YES, 316SS	(1)
	32	Manufacture	As per AML / PML	(2)
	33	Model No.	By VENDOR	(3)

Notes:

1. Vendor to provide 316 SS nameplate or tag number permanently attached to instrument.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. VENDOR shall submit proper model number on bid proposal.
4. VENDOR shall provide a complete set of accessories, mounting and installation devices and special tools (if any).
5. VENDOR Shall provide inline Resistor and End of line Resistor for open loop/short circuit monitoring.
6. To be specified and finalized during detail engineering.
7. Vendor to supply special tools for calibration.

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR MANUAL CALL POINT (OUTDOOR)	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 16 dari 21

Tag Number	: See List	Location	: See List
Drawing No.	: -	Equip./Line No.	: -

GENERAL	1	Tag No.	See List	
	2	Application	Manual Initiation of Fire Alarm	
	3	Area Classification	Zone 2, Group HA, T3	
	4	Enclosure Class	IP65 / Outdoor	
	5	Mounting	Surface	
	6	Enclosure Material	Aluminum copper free	(4)
	7	Quantity		
	8			
SPECIFICATION	9	Type	Break glass	
	10	Enclosure Colour	Red	
	11	Push Button Colour	Red or Manufacture Standard	
	12	Protective Cover	Required	
	13	Contact Form	2 (two) SPDT (VTA)	
	14	Power Supply	24 VDC @ 1 Amp	
	15	Cable Entry	½" NPTF	
	16	Connection	Screwed	
	17	Wiring / Termination	2 wires	
	18			
ENVIRONMENT SPECIFICATION	19	Ambient Temperature	27.2 °C (min. 25.5 °C - max. 29 °C)	
	20	Relative Humidity	82.1 % (min. 74% - max. 90.3%)	
	21	Rainfall	14.9 mm (min. 0.5 mm - max. 64.8 mm)	
	22	Velocity	2.11 m/s (min. 1.5 m/s - max. 3.2 m/s)	
	23	Barometer	1 atm	
	24			
OPTIONS	25	Mounting Bracket	YES	(4)
	26	Cover	Cover shall be provided to prevent mis-operation	
	27	LED Indicator	N/A	
	28	Resistor	YES	(5)
	29	Reset Key	YES	
	30			
SELECTION	31	Name Plate (Tagging Number)	YES, 316SS	(1)
	32	Manufacture	As per AML / PML	(2)
	33	Model No.	By VENDOR	(3)

Notes:

1. Vendor to provide 316 SS nameplate or tag number permanently attached to instrument.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. VENDOR shall submit proper model number on bid proposal.
4. VENDOR shall provide a complete set of accessories, mounting and installation devices and special tools (if any).
5. VENDOR Shall provide inline Resistor and End of line Resistor for open loop/short circuit monitoring.
6. To be specified and finalized during detail engineering.
7. Vendor to supply special tools for calibration.

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG
DATA SHEET FOR MANUAL CALL POINT	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 17 dari 21	

MANUAL ALARM CALL POINT LIST

No.	Tag Number	Location	Service	P&ID No.	REMARK
1	MACP-001	INDOOR	ELECTRICAL PANEL ROOM	TMBP-SP001-IN-LAY-D109	
2	MACP-002	INDOOR	INSTRUMENT PANEL ROOM	TMBP-SP001-IN-LAY-D109	
3	MACP-003	INDOOR	OPERATOR ROOM	TMBP-SP001-IN-LAY-D109	
4	MACP-004	INDOOR	FRONT OFFICE & LOBBY AREA	TMBP-SP001-IN-LAY-D109	
5	MACP-005	INDOOR	PAINTRY	TMBP-SP001-IN-LAY-D109	
6	MACP-001	OUTDOOR	PIG RECEIVER	TMBP-SP001-IN-LAY-D108	
7	MACP-002	OUTDOOR	WATER DISTRIBUTION PUMP (P-017 A/B)	TMBP-SP001-IN-LAY-D108	
8	MACP-003	OUTDOOR	GROSS TRANSFER PUMP (P-001 A/B)	TMBP-SP001-IN-LAY-D108	

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR HORN BEACON (INDOOR)	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 18 dari 21

Tag Number	: See List	Location	: See List
P&ID No.	: -	Equip./Line No.	: -

GENERAL	1	Quantity	See List
	2	Location	See List
	3	Protection Degree of Enclosure	IP 42 (Indoor)
	4	Construction	High Impact Flame Retardant Plastic
	5	Applicable Standard	UL / FM / CSA
	6	Factory & Site Test	Yes, Required
	7	Area Classification	Non Hazardous/Safe Area
	8		
DESIGN DATA	9	Type	Electronic Sounder & Strobe Beacon
	10	Tone Frequency	Min. 28 Tone Selectable (4)
	11	Volume Control	Integral
	12	Case Material	Corrosion-free GRP
	13	Case Color	Red (for Fire) & Blue (for Gas)
	14	Sound Output	Minimum 100 dB at 1 meter from the source
	15	Strobe Output	Red Light with effective intensity of 100 Cd
	16	Flash Rate	1 Hz
	17	Power Supply	24 VDC Loop Power (Max. 48 Watt (DC)) or External power 230 VAC
	18	Maximum Power Consumption	Manufature Standard (Note 4)
	19	Electrical Connection	½" NPTF
	20	Cable Size	14 or 16 AWG (1 P-2.5 mm^2)
	21		
ENVIRONMENT SPECIFICATION	22	Ambient Temperature	27.2 °C (min. 25.5 °C - max. 29 °C)
	23	Relative Humidity	82.1 % (min. 74% - max. 90.3%)
	24	Rainfall	14.9 mm (min. 0.5 mm - max. 64.8 mm)
	25	Velocity	2.11 m/s (min. 1.5 m/s - max. 3.2 m/s)
	26	Barometer	1 atm
	27		
OTHERS	28	Mounting Bracket c/w with bolt & nuts	Yes, Required
	29	Spare Part 2 Years	Yes, Required
	30	End of Line Resistor	Yes, Required
	31		
REMARKS	32	Name Plate (Tagging Number)	YES, 316SS (1)
	33	Manufacture	As per AML / PML (2)
	34	Model No.	By VENDOR (3)

Notes:

1. Vendor to provide 316 SS nameplate or tag number permanently attached to instrument.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. VENDOR shall submit proper model number on bid proposal.
4. As per Manufacture Standard
5. The tone of fire/gas/general alarm shall be different than abandon alarm tone
   When abandon alarm is triggered, all hom (fire/gas/g eneral alarm) will silmutaneously active
   and have the same tone with the abandon alarm tone
6. To be specified and finalized during detail engineering.

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR HORN BEACON (OUTDOOR)	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 19 dari 21

Tag Number	: See List	Location	: See List
Drawing No.	: -	Equip./Line No.	: -

GENERAL	1	Quantity	See List
	2	Location	See List
	3	Protection Degree of Enclosure	Min. IP65 & Weather Proof
	4	Construction	High Impact Flame Retardant Plastic
	5	Applicable Standard	UL / FM / CSA
	6	Factory & Site Test	Yes, Required
	7	Area Classification	Zone 2, Group IIA, T3
	8		
DESIGN DATA	9	Type	Electronic Sounder & Strobe Beacon
	10	Tone Frequency	Min. 28 Tone Selectable (4)
	11	Volume Control	Integral
	12	Case Material	Corrosion-free GRP
	13	Case Color	Red (for Fire) & Blue (for Gas)
	14	Sound Output	Minimum 100 dB at 1 meter from the source
	15	Strobe Output	Red Light with effective intensity of 100 Cd
	16	Flash Rate	1 Hz
	17	Power Supply	24 VDC Loop Power (Max. 48 Watt (DC)) or External power 230 VAC
	18	Maximum Power Consumption	Manufature Standard (Note 4)
	19	Electrical Connection	½" NPTF
	20	Cable Size	14 or 16 AWG (1 P-2.5 mm^2)
	21		
ENVIRONMENT SPECIFICATION	22	Ambient Temperature	27.2 °C (min. 25.5 °C - max. 29 °C)
	23	Relative Humidity	82.1 % (min. 74% - max. 90.3%)
	24	Rainfall	14.9 mm (min. 0.5 mm - max. 64.8 mm)
	25	Velocity	2.11 m/s (min. 1.5 m/s - max. 3.2 m/s)
	26	Barometer	1 atm
	27		
OTHERS	28	Mounting Bracket c/w with bolt & nuts	Yes, Required
	29	Spare Part 2 Years	Yes, Required
	30	End of Line Resistor	Yes, Required
	31		
REMARKS	32	Name Plate (Tagging Number)	YES, 316SS (1)
	33	Manufacture	As per AML / PML (2)
	34	Model No.	By VENDOR (3)

Notes:

1. Vendor to provide 316 SS nameplate or tag number permanently attached to instrument.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. VENDOR shall submit proper model number on bid proposal.
4. As per Manufacture Standard
5. The tone of fire/gas/general alarm shall be different than abandon alarm tone
   When abandon alarm is triggered, all hom (fire/gas/g eneral alarm) will silmutaneously active
   and have the same tone with the abandon alarm tone
6. To be specified and finalized during detail engineering.

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG
DATA SHEET FOR HORN BEACON	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 20 dari 21	

HORN BEACON LIST

No.	Tag Number	Location	Service	P&ID No.	REMARK
1	HRN-001	INDOOR	ELECTRICAL PANEL ROOM	TMBP-01-IN-LAY-109	
2	HRN-002	INDOOR	INSTRUMENT PANEL ROOM	TMBP-01-IN-LAY-109	
3	HRN-003	INDOOR	OPERATOR ROOM	TMBP-01-IN-LAY-109	
4	HRN-004	INDOOR	FRONT OFFICE & LOBBY AREA	TMBP-01-IN-LAY-109	
5	HRN-005	INDOOR	PAINTRY	TMBP-01-IN-LAY-109	
6	BCN-001	OUTDOOR	PIG RECEIVER	TMBP-01-IN-LAY-108	
7	BCN-002	OUTDOOR	WATER DISTRIBUTION PUMP (P-017 A/B)	TMBP-01-IN-LAY-108	
8	BCN-003	OUTDOOR	GROSS TRANSFER PUMP (P-001 A/B)	TMBP-01-IN-LAY-108	

skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR FIRE ALARM CONTROL PANEL (FACP)	No. Dok. : TMBP-SP001-IN-DST-D115 Rev. : A Halaman : 21 dari 21

Tag Number	: See List	Service	: -
Drawing No.	: -	Location	: Control Room

GENERAL	1	Quantity	1 EA
	2	Hazardous Area Classification	Non Hazardous Area
	3	Enclosure Protection Type	IP42
	4	Required Certification	UL, FM
	5	Installation	Indoor c/w Air Conditioning
	6		
PANEL SPECIFICATION	7	Type	Full Addressable
	8	Power Supply	230 VAC, 50 Hz
	9	Battery Type	Dry Battery
	10	Capacity	Min. 2 hr or Manufactured Standard
	11	Charger	Built in Panel
	12	LCD Display	YES
	13	Keypad	YES
	14	I/O Module	YES (at terminal Box Module)
	15	Enclosure Material	Steel Galvanize, 2 mm thickness minimum
	16	Enclosure Colour	Red, RAL standard
	17	Enclosure Mounting	Wall Mounting c/w Accessories
	18	Enclosure Connection	½" NPT
	19	Size Dimension	Manufactured Standard
	20	Weight	Manufactured Standard
	21	Output & Communication	Hardwire & Modbus
	22		
SYSTEM FEATURES/ CAPABILITIES	23	Fire Alarm Indication	YES, required
	24	Activation Alarm Buzzer/Hom	YES, required (100 dBA minimum at 1 meter distance)
	25	Visual Alarm	Zone Fire Alarm or Pre-alarm of Manual Controller Actuation
	26		Inergen Pre-Discharge & Discharge Alarm
	27		Common Alarm
	28		Fault Alarm
	29	Fault Alarm	Battery loss
	30		AC Power Loss
	31		Charger Failure
	32		Trouble Line Circuit (Open/Short)
	33		
INTERFACE TO OTHER SYSTEM	34	Auxiliary Panel	N/A
	35	Emergency Shutdown System	N/A
	36	Fire and Gas System	FM-200 (Electrical Room Power Plant Area)
	37		
REMARKS	38	Spare Part 2 Years	YES, required
	39	Name Plate (Tagging Number)	YES, 316SS (1)
	40	Manufacture	As per AML / PML (2)
	41	Model No.	By VENDOR (3)

Notes:

1. All instrument shall be tagged with 316 SS nameplate showing tag no.
2. VENDOR subject to shall be refer to PT. PERTAMINA HULU ROKAN Preferred Manufacturer List (PML)
3. Vendor shall submit proper model number on bid proposal.
4. As per manufacture standard
5. Able to deliver power & Logic to Selonoid Valve 24 VDC